

Cheat Sheet: Recognize the Integral at a Glance

A fast lookup table for exam review. Pair this with “Which Integration Technique Do I Use?” if you want the full step-by-step reasoning behind any row.

If the integral looks like...	Use...	Quick example
x^n (plain power of x)	Power Rule	$\int x^3 dx$
$\sin x, \cos x, e^x, \frac{1}{x}$ (basic function, plain x inside)	Basic formula, direct	$\int e^x dx$
$f(g(x)) \cdot g'(x)$ — inner function’s derivative sitting right there	u -substitution	$\int 2x(x^2 + 1)^3 dx$
Numerator = derivative of denominator	Quick ln shortcut (a special case of substitution)	$\int \frac{2x-6}{x^2-6x+8} dx$
Polynomial $\times \sin x, \cos x$, or e^x	Integration by Parts	$\int x \sin x dx$
Polynomial $\times \ln x$, or $\ln x$ alone	Integration by Parts	$\int x^5 \ln x dx$
$e^x \sin x$ or $e^x \cos x$	Integration by Parts, twice, then solve algebraically	$\int e^x \cos x dx$
$\sin^m x \cos^n x$, same angle, at least one power odd	Trig Integral — peel off one factor, use $\sin^2 + \cos^2 = 1$	$\int \sin^3 x \cos^2 x dx$
$\sin^m x \cos^n x$, same angle, both powers even	Trig Integral — power-reducing identities	$\int \sin^2 x \cos^2 x dx$
$\sqrt{1 \pm \cos(2x)}$	Double-angle identity eliminates the root directly	$\int \sqrt{1 + \cos 2x} dx$
$\sqrt{a^2 - x^2}, \sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$, nothing else going on	Basic arcsin/arctan formula (or complete the square first)	$\int \frac{dx}{\sqrt{4 - x^2}}$
Same roots as above, but with extra polynomial factors or powers	Trig Substitution	$\int x^2 \sqrt{9 - x^2} dx$
$\frac{P(x)}{Q(x)}$, factorable denominator, proper fraction	Partial Fractions	$\int \frac{2x+3}{(x-1)(x+2)} dx$
$\frac{P(x)}{Q(x)}$, numerator degree $\geq$ denominator degree	Long division first, then handle the remainder	$\int \frac{x^3 + 3x^2 + 2x - 7}{x + 2} dx$
An ∞ bound, or a vertical asymptote inside the interval	Improper Integral — replace the trouble spot with a limit	$\int_1^\infty \frac{dx}{x^2}$

The single most useful habit

Before matching a row above, always ask: can I simplify this algebraically first? Expanding a product, dividing a fraction term-by-term, or applying a trig identity often turns a “hard-looking” integral into a one-line power-rule problem. Many of the trickiest-looking integrals in this course collapse dramatically once you simplify before hunting for a technique.